Project Synopsis — Resume Screening System

HCLTech Career Shaper Programme | Project No. 14 | Academic Year 2025-26

Field	Project Details
Project Title	AI-Based Resume Screening and Candidate Shortlisting System
Project No.	14
Domain	Digital-Data-AI / AIML
Difficulty	Advanced
Duration	4 Weeks
Format	Individual Project
Learner Name	Vivek Yadav
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College / Campus	Abes Engineering College
Mentor Name	Natesan Rajan
Submission Date	29-Oct-2026

1. Problem Statement

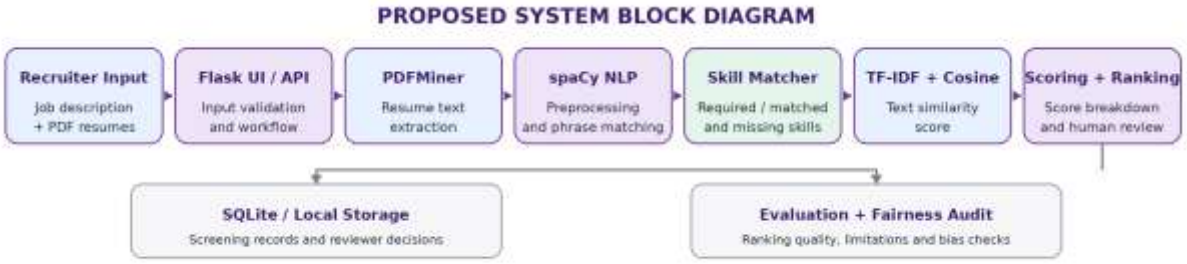
Recruiters often spend considerable time manually reviewing resumes against each job description, while resume formats and terminology vary across candidates. This project proposes a decision-support application that extracts relevant information from PDF resumes, compares it with job requirements, and presents explainable candidate scores and rankings. The system will support human review rather than make hiring decisions autonomously, and its performance, privacy risks, and potential bias will be assessed.

2. Proposed Solution & Approach

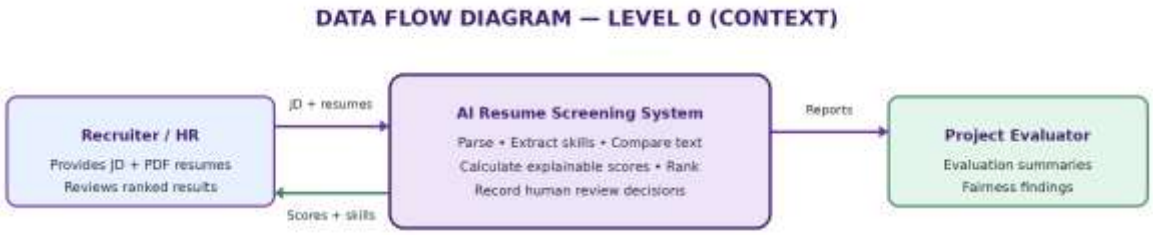
The proposed prototype will use Flask for the application workflow, pdfminer.six for PDF text extraction, and spaCy for text preprocessing and skill-phrase identification. A configurable matching engine will use a domain-specific skill dictionary and scikit-learn TF-IDF with cosine similarity to compare resume content with the job description. An explainable scoring module is planned to combine text similarity, required-skill coverage, relevant experience, and education match. The recruiter interface will show matched and missing skills, component scores, ranked candidates, and a separate human-review decision; evaluation and fairness findings will be documented after implementation.

3. System Architecture

Proposed system block diagram:



Data Flow Diagram (DFD) — Level 0:



4. Technology Stack

Frontend / UI: HTML, CSS, Bootstrap, JavaScript (proposed).
Backend / Logic: Python, Flask, REST API endpoints.
Database: SQLite for local prototype persistence.
AI/ML Framework: spaCy; scikit-learn (TF-IDF, cosine similarity and evaluation metrics).
Visualisation: Pandas, Matplotlib and Seaborn.
Hardware: Standard laptop/PC; no dedicated GPU is expected for the baseline design.

5. Dataset Used

Dataset file: dataset.csv (provided with the project).
Current size: 500 rows × 8 columns; initial inspection found no missing values.
Job roles present: 5 — Data_Analyst (102), Software_Engineer (87), HR_Executive (120), QA_Engineer (104), Project_Manager (87).
Key columns: job_role, years_exp, skills_matched, keyword_density, shortlisted.
Label distribution: 173 shortlisted (34.6%) and 327 not shortlisted (65.4%).
Additional data: No additional real-world data collected at this stage.
Data note: data_dictionary.csv still contains placeholder descriptions. The supplied predicted_score has near-zero linear correlation with shortlisted (r ≈ -0.009); label logic must be checked before supervised model training.

6. Key Results & Metrics

Preliminary data completeness: 0 missing values in the supplied 500 × 8 CSV at initial load; this is not a model-performance metric.
Current label balance: 173/500 shortlisted (34.6%); 327/500 not shortlisted (65.4%).
Matching quality / response time: Not yet measured. Precision@K or NDCG@K and average processing time will be measured on a manually reviewed synthetic resume–JD test set.
Classification metrics: Accuracy, precision, recall and F1 will be reported only if a valid labeled test set is established.
Expected business value: Provide consistent, explainable initial screening support and help recruiters inspect relevant skills faster; business impact has not yet been measured.

7. Challenges Faced & How Resolved

The following are anticipated implementation risks and planned mitigations; update this section with actual challenges and outcomes after development.
Challenge 1 — PDF resumes may have inconsistent layouts or unreadable text. Mitigation: validate extracted text, handle parsing failures, and flag uncertain records for human review.
Challenge 2 — Existing shortlist labels may not represent reliable outcomes, and demographic cues may affect results. Mitigation: audit label-generation logic, test on manually reviewed pairs, run controlled counterfactual fairness checks, and retain human decision authority.

8. References

1. HCLTech Career Shaper Programme. III Year Industry Project — Project Brief, Project No. 34 (provided course document, 2025-26).
2. spaCy. “Language Processing Pipelines.” https://spacy.io/usage/processing-pipelines/ (accessed 29 October 2026).
3. scikit-learn. “TfidfVectorizer.” https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html (accessed 29 October 2026).
4. scikit-learn. “cosine_similarity.” https://scikit-learn.org/stable/modules/generated/sklearn.metrics.pairwise.cosine_similarity.html (accessed 29 October 2026).
5. pdfminer.six. Documentation. https://pdfminersix.readthedocs.io/en/latest/ (accessed 29 October 2026).
6. Ministry of Electronics and Information Technology, Government of India. Digital Personal Data Protection Rules, 2025. https://www.meity.gov.in/documents/act-and-policies/digital-personal-data-protection-rules-2025-gDOxUjMtQWa (accessed 29 October 2026).

9. Ethical Compliance Declaration

Draft declaration — complete only after checking each statement against the implemented project and applicable programme requirements.

I confirm that:

- All dataset usage complies with the source license.
- No personally identifiable information (PII) is stored without consent.
- AI tools used for code/writing are disclosed in the Appendix.
- This project complies with the DPDP Act 2023 where applicable.

Learner Signature: Vivek Yadav

Date: 29 October 2026